Fiscal Year:	FY 2025	**Task Last Updated:**	FY 11/06/2025
PI Name:	Williams, Michael A. M.D.		
Project Title:	Pilot Feasibility Study of Safety and Effectiveness of Telemetric Intracranial Pressure Monitoring in Astronauts Before, During, and After Commercial Orbital Spaceflight		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	(1) **SANS**:Risk of Spaceflight Associated Neuro-ocular Syndrome (SANS)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	maw99@uw.edu	**Fax:**	FY 206-744-9942
PI Organization Type:	UNIVERSITY	**Phone:**	206-616-7291
Organization Name:	University of Washington		
PI Address 1:	Box 359924		
PI Address 2:	325 9th Avenue		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98104-2499	**Congressional District:**	7
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	11/01/2022	**End Date:**	02/29/2024
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-EXP0014		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Spaceflight Associated Neuro-ocular Syndrome (SANS) is a disorder after long-duration spaceflight characterized by swelling of the optic nerve (papilledema) folds in the choroid layer of the eye, flattening of the optic globe of the eye, and change in near vision (Mader, 2011). Up to 70% of astronauts develop SANS, which is considered a “red risk” – because if severe visual impairment developed, astronauts’ ability to function safely or to control a spacecraft could be compromised (Patel ZS, 2020). SANS is thought to be caused by abnormal intracranial pressure (ICP); however non-invasive ICP methods researched by NASA have not been accurate enough to make this determination (Stenger MR, 2017). Lumbar punctures are considered too risky to perform during spaceflight (Barr YR, 2014). The underlying cause of SANS remains unknown, and, despite evidence during parabolic flight that ICP may not be pathologically elevated during brief periods of weightlessness (Lawley, 2017), measures of ICP during long-duration spaceflight are needed to understand whether abnormal ICP plays a role in the development of ocular and brain structural changes. Accordingly,		

	this proposal is relevant to mitigating the gap SANS-102: Determine the relationship between the fluid shifts induced ocular changes and fluid shifts in the central nervous system (CNS), including whether elevated intracranial pressure or brain edema play a role.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	IMPACT: The successful completion of the FDA QSub and IDE EFS applications provides regulatory approval for surgical implantation of the Miethke M.scio telemetric ICP sensor, which will demonstrate that ICP can be measured accurately and safely before, during, and after spaceflight. If the pilot early feasibility study is successful, then future research using the same FDA IDE EFS approval can be submitted to the FDA.
Task Progress:	KEY FINDINGS September 11, 2023: IDE Study Suspension Supplement G230002/S002 was received by the FDA. October 4, 2023: Letter received from FDA indicating that the Study Suspension Supplement had been reviewed and that no further information was required. The FDA requires submission of an IDE supplement requesting approval if the study is to be resumed. November 20, 2023: Root Cause Analysis report, as incorporated into the FDA was finalized and sent to 3 peer reviewers. January 17, 2024: RCA with modifications based on comments by the peer reviewers, was submitted to the TRISH, and a copy to SpaceX. The RCA recommendations can be used to guide future research using this approach. The FDA EFS IDE remains active with the PI as sponsor-investigator. Because the project was ended, no future research is currently planned.
Bibliography Type:	Description: (Last Updated: 05/15/2026)
Articles in Peer-reviewed Journals	Eklund A, Bäcklund T, Williams MA, Jan M. "Telemetric lumbar cerebrospinal fluid pressure monitoring for spaceflight: A novel approach to spaceflight-associated neuro-ocular syndrome research." npj Microgravity. 2026 May 7. Early access article. https://doi.org/10.1038/s41526-026-00586-0 ; PMID: 42120394 , May-2026